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OnDeBERTa: An Ordered Neuron Memory Enhanced DeBERTa Model for E-commerce Review Sentiment Classification

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***Abstract*—This paper proposes OnDeBERTa, an enhanced DeBERTa-based model for e-commerce review sentiment classification. The proposed model preserves the DeBERTa encoder and introduces a lightweight ON-LSTM decoder together with an attention pooling mechanism to improve the modeling of negation scope, contrastive structures, and long-distance sentiment cues. Using the Amazon Review Polarity dataset, this study compares traditional machine learning models, pretrained language models, and DeBERTa-based decoder-augmented models under a unified experimental protocol. Experimental results show that the one-layer OnDeBERTa achieves 94.18% Accuracy, 92.26% Precision, 91.02% Recall, and 91.64% F1-score, outperforming DeBERTa-base, DeBERTa + LSTM, and DeBERTa + BiLSTM. The ablation study shows that attention pooling is the most effective aggregation strategy, while increasing ON-LSTM depth further improves model performance. A qualitative attention analysis further indicates that OnDeBERTa focuses more accurately on contrastive cues and decisive negative sentiment fragments. These results suggest that the performance gain mainly comes from the integration of DeBERTa contextual encoding, ON-LSTM hierarchical memory modeling, and attention-based sentiment aggregation.**

Keywords—DeBERTa, ON-LSTM, Sentiment Classification, E-commerce Reviews, OnDeBERTa.

I. INTRODUCTION

E-commerce reviews reflect customers' opinions on product quality, delivery, after-sales service, and overall shopping experience. Accurate sentiment polarity classification helps merchants identify negative feedback, improve services, and support business decision-making. Therefore, e-commerce review sentiment classification is an important task in natural language processing and intelligent user feedback analysis for e-commerce platforms [1], [2].

Traditional sentiment classification methods usually use bag-of-words, TF-IDF, or n-gram features with classifiers such as Naive Bayes, Logistic Regression, and Support Vector Machines [3]. Although these methods are efficient and easy to implement, they rely heavily on sparse lexical features and have limited ability to model word order, contextual dependencies, and semantic composition. As a result, they may perform poorly when reviews contain negation, contrast, or mixed sentiment expressions.

Pretrained language models, such as BERT, RoBERTa, and DeBERTa, have significantly improved text classification performance [4] – [6]. DeBERTa further enhances contextual representation by modeling content and relative position information separately through disentangled attention. This makes it effective for capturing order-sensitive sentiment cues in e-commerce reviews. However, standard DeBERTa-based classifiers usually rely on the [CLS] representation or simple pooling, which may not sufficiently highlight key sentiment fragments in complex reviews.

To address this limitation, this paper proposes OnDeBERTa, an enhanced DeBERTa-based model for e-commerce review sentiment classification [6], [7]. Unlike common DeBERTa-based hybrid models that append standard LSTM or BiLSTM layers, OnDeBERTa introduces ON-LSTM to impose an explicit hierarchical memory structure on DeBERTa token representations. Together with attention pooling, the proposed model aims to better aggregate key sentiment fragments while preserving the original DeBERTa encoder structure.

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(1) An enhanced DeBERTa-based sentiment classification model, OnDeBERTa, is proposed to improve the modeling of hierarchical structures, negation scope, and long-distance sentiment cues in complex e-commerce reviews.

(2) A unified experimental comparison is conducted on the Amazon Review Polarity dataset, covering traditional machine learning methods, pretrained language models, and sequence-enhanced DeBERTa variants.

(3) Ablation experiments are performed to analyze the effects of pooling strategies and ON-LSTM depth, demonstrating the contribution of hierarchical memory to sentiment polarity classification.

II. Related Work

A. Traditional Machine Learning Methods

Traditional sentiment classification methods usually combine TF-IDF, bag-of-words, or n-gram features with classifiers such as Naive Bayes, Logistic Regression, and Support Vector Machines [1]. These methods are simple, efficient, and often used as text classification baselines. However, they mainly rely on sparse lexical features and have limited ability to capture word order, contextual dependencies, and semantic composition. Therefore, their performance is often limited when dealing with long reviews, negation, contrastive structures, and mixed sentiment expressions.

B. Pretrained Language Models

Pretrained language models learn contextual representations from large-scale corpora and have become mainstream methods for text classification [4], [5]. BERT uses a bidirectional Transformer encoder to capture contextual semantics [4], while RoBERTa improves the pretraining strategy [5]. DeBERTa further separates content and positional information through disentangled attention, enhancing its ability to model relative positions and semantic interactions [6]. In e-commerce review sentiment classification, these models can better handle long reviews and complex expressions [7], [9]. However, standard classification heads usually rely on [CLS] vectors or simple pooling, which may not fully capture hierarchical structures and key sentiment fragments in reviews.

C. ON-LSTM Sequence Modeling

LSTM addresses long-term dependency problems through gated mechanisms, and BiLSTM extends this ability by modeling both forward and backward contexts. However, standard LSTM and BiLSTM mainly capture linear sequence relationships and lack an explicit hierarchical inductive bias. ON-LSTM introduces ordered neurons and the cumax master gate mechanism to model hierarchical structures [7]. In this design, high-level neurons preserve long-term semantic information, while low-level neurons capture local token changes. This makes ON-LSTM suitable for modeling negation scope, contrastive structures, and delayed sentiment evidence in sentiment classification [9].

III. Proposed Method

A. Overall Architecture

The proposed OnDeBERTa model is designed as a sequential encoding – decoding – classification framework, as shown in Fig. 1. It consists of four components: a DeBERTa encoder, an ON-LSTM decoder, an attention pooling layer, and a classification layer. The review title and content are first concatenated and tokenized by the DeBERTa tokenizer. The resulting sequence is then encoded by DeBERTa to obtain contextual token representations. Although LSTM and BiLSTM are commonly used to capture linear sequential dependencies, their ability to model hierarchical sentence structures is relatively limited. To address this limitation, OnDeBERTa introduces the ordered-neuron memory mechanism of ON-LSTM, enabling hierarchical sequential modeling over DeBERTa representations. Finally, attention pooling aggregates informative token representations into a review-level vector, which is used for final sentiment polarity prediction.

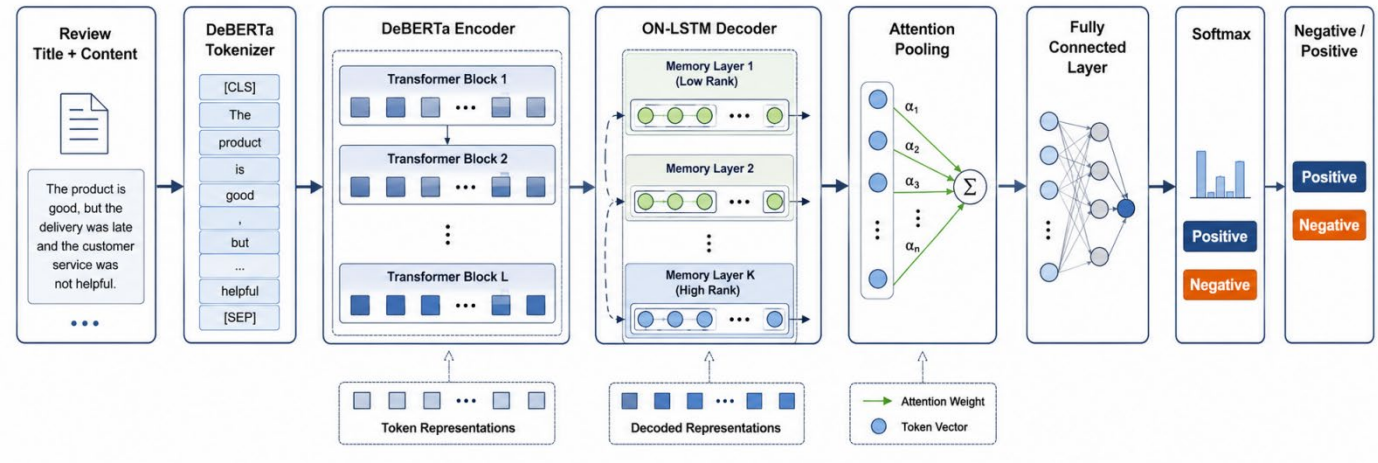


Fig. 1. Overall Architecture of the OnDeBERTa Model

The main idea of OnDeBERTa is to combine the contextual modeling ability of DeBERTa with the hierarchical memory mechanism of ON-LSTM. DeBERTa captures global semantic dependencies through self-attention, while ON-LSTM further models ordered and hierarchical temporal information in the encoded sequence. This design is particularly suitable for e-

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commerce reviews, where sentiment polarity is often determined by negation expressions, contrastive clauses, and long-distance dependencies.

B. DeBERTa-based Contextual Encoding

In the DeBERTa-based contextual encoding module, the input review is first transformed into a token sequence for semantic representation learning. Let the input review sequence be denoted as:

$$X = [x_1, x_2, \dots, x_n] \quad (1)$$

where N is the sequence length, and x_t represents the t-th input token. The input sequence is fed into the DeBERTa encoder to generate contextual token representations:

$$H = DeBERTa(X) = [h_1, h_2, \dots, h_n], h_t \in R^d \quad (2)$$

where h_t denotes the contextual representation of the t-th token.

Different from standard Transformer encoders, DeBERTa adopts a disentangled attention mechanism to model content information and relative position information separately. This enables the model to better capture semantic relations and word-order-sensitive expressions in review texts. For example, in "The product looks good, but the delivery was late," the clause after "but" plays a more important role in determining the final sentiment. Therefore, DeBERTa provides strong contextual representations for subsequent hierarchical sequence modeling.

C. ON-LSTM Hierarchical Sequence Decoding

The contextual representations generated by DeBERTa are then passed into the ON-LSTM decoder:

$$Z = ONLSTM(H) = [z_1, z_2, \dots, z_n] \quad (3)$$

where z_t denotes the decoded representation of the t-th token.

Compared with standard LSTM, ON-LSTM introduces ordered neurons and master gates to model hierarchical structures in sequential data. Its key operation is the cumax function:

$$\text{cumax}(x)_k = \sum_{j=1}^{k} \text{softmax}(x)_j, k = 1, \dots, d \quad (4)$$

Based on this mechanism, ON-LSTM controls the update and retention of information at different neuron levels. Low-level neurons tend to capture local token-level changes, while high-level neurons preserve longer-term semantic information. This helps the model handle sentiment expressions involving negation, contrast, and delayed key evidence.

D. Attention Pooling and Classification

To generate the final review-level representation, this study considers three pooling strategies: Last Pooling, Mean Pooling, and Attention Pooling. Last Pooling uses the final hidden state as the sequence representation. Mean Pooling averages all token representations. Attention Pooling assigns different weights to tokens according to their importance for sentiment classification, allowing the model to focus more on key sentiment cues.

The core computation of Attention Pooling is defined as:

$$s = \Sigma_{t=1}^{N} \alpha_t z_t \quad (5)$$

where α_t denotes the attention weight of the t-th token, z_t is the corresponding decoded token representation, and s is the final review-level vector.

Finally, s is fed into a fully connected layer and a Softmax classifier to predict the sentiment polarity:

$$\hat{y} = \text{softmax}(W_s s + b_s) \quad (6)$$

where $\hat{y}$ represents the predicted probability distribution over the two sentiment classes: Negative and Positive.

IV. Experiments

A. Dataset

This paper uses the Amazon Review Polarity dataset for experiments [10]. The dataset is derived from Amazon product reviews and is a commonly used binary classification dataset in e-commerce review sentiment classification [11]. The full dataset contains 3,600,000 training samples and 400,000 testing samples. In the training set, there are 1,800,000 positive samples and 1,800,000 negative samples. In the testing set, there are 200,000 positive samples and 200,000 negative samples. Each review text consists of a review title and review content, which can effectively reflect product evaluation, logistics experience, service feedback, and overall purchasing sentiment in real e-commerce scenarios.

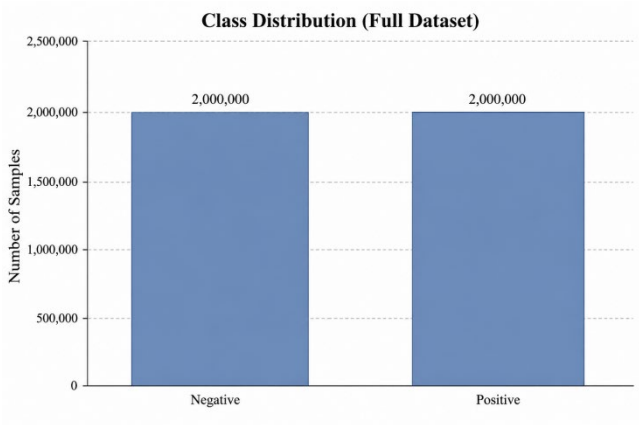


Fig. 2. Class Distribution of the Full Amazon Review Polarity Dataset

As shown in Fig 2, the number of Negative and Positive samples is the same, indicating that the dataset is well balanced at the class level. This characteristic helps reduce the influence of class bias during model training and provides a fair basis for comparing different models. In addition to class distribution, review length is also an important factor affecting model performance. As shown in Fig. 3, the review length distribution presents a clear long-tail pattern after concatenating the review title and content. Most reviews are concentrated in short and medium-length ranges, while a smaller proportion extends to nearly 1000 characters. This distribution indicates that the dataset contains texts of varying lengths, making it suitable for evaluating the model's ability to capture both local sentiment cues and long-distance dependencies.

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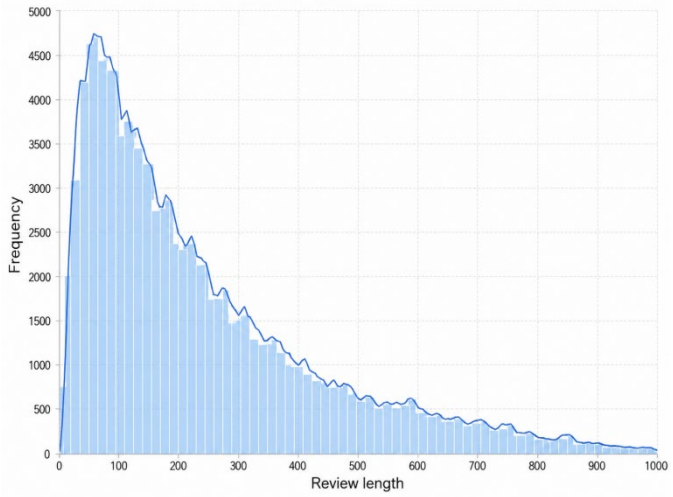


Fig. 3. Review Length Distribution of the Dataset

B. Evaluation Metrics and Experimental Setup

This study evaluates model performance using four standard metrics: Accuracy, Precision, Recall, and F1-score. Accuracy measures the overall classification correctness, while Precision and Recall evaluate the reliability and coverage of positive predictions. F1-score provides a balanced measure by combining Precision and Recall. Since Amazon Review Polarity is a balanced binary sentiment classification dataset, these metrics are suitable for comparing different models.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (7)$$

$$Precision = \frac{TP}{TP+FP} \quad (8)$$

$$Recall = \frac{TP}{TP+FN} \quad (9)$$

$$F1-score = 2 \times \frac{Precision \times Recall}{Precision+Recall} \quad (10)$$

All experiments were conducted on a single workstation equipped with an NVIDIA A100 GPU. Deep models, including BERT-base, RoBERTa-base, DeBERTa-base, DeBERTa + LSTM, DeBERTa + BiLSTM, and the proposed OnDeBERTa, were trained with AdamW using a learning rate of 2e-5, batch size of 32, dropout rate of 0.10, and maximum sequence length of 256. The maximum number of training epochs was set to 50, with early stopping based on the validation F1-score. The official full Amazon Review Polarity dataset was used, containing 3,600,000 training reviews and 400,000 testing reviews with balanced positive and negative classes. A validation set was separated from the official training split for model selection, while the official testing set was used only for final evaluation.

C. Comparative Experiments

We conduct a comparative evaluation of all models listed in Table I on the Amazon Review Polarity test set under a unified experimental protocol. Beyond comparing classification accuracy, this experiment aims to verify whether hierarchical sequence modeling can further enhance DeBERTa in understanding complex e-commerce reviews. Such reviews often contain negation, contrastive clauses, and delayed sentiment evidence, where the final polarity depends on the relationship between multiple text segments. Therefore, the comparison focuses on whether ON-LSTM can provide additional hierarchical memory over DeBERTa representations and whether attention pooling can better aggregate key sentiment fragments for final prediction. Classical machine learning models use TF-IDF features with default settings, while FastText uses word embeddings for text representation. BERT-base, RoBERTa-base, DeBERTa-base, and the DeBERTa-based decoder-augmented models are trained following the experimental setup described above. For fair comparison, DeBERTa + LSTM, DeBERTa + BiLSTM, and the main OnDeBERTa model all use a one-layer decoder with attention pooling. This setting allows the effect of ON-LSTM to be directly compared with standard recurrent decoders under the same decoder depth. Unless otherwise specified, the remaining hyperparameters follow the global training settings.

TABLE I. OVERALL PERFORMANCE COMPARISON OF ONDEBERTA AND BASELINE MODELS ON THE DATASET

Model	Hyperparameters	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Multinomial Naive Bayes	TF-IDF; defaults	84.72	78.65	72.40	75.39
Logistic Regression	TF-IDF; defaults	87.96	83.10	80.36	81.71
SVM	TF-IDF; defaults	89.12	84.76	82.31	83.52
FastText	word embedding; defaults	90.48	86.72	84.95	85.83
BERT-base	global settings	91.62	88.91	86.74	87.81
RoBERTa-base	global settings	92.31	89.75	88.08	88.91
DeBERTa-base	global settings	93.05	90.68	89.16	89.91
DeBERTa + LSTM	1-layer decoder; attention pooling	93.31	91.05	89.64	90.34
DeBERTa + BiLSTM	1-layer decoder; attention pooling	93.58	91.40	90.12	90.76
OnDeBERTa	1-layer ON-LSTM; attention pooling	**94.18**	**92.26**	**91.02**	**91.64**

As shown in Table I, pretrained language models generally outperform traditional machine learning methods, indicating the advantage of contextual representations over sparse lexical features. Among the encoder-only models, DeBERTa-base achieves better performance than BERT-base and RoBERTa-base, showing its stronger ability to model contextual and

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position-sensitive sentiment information. Based on DeBERTa, the decoder-augmented models further improve the results. Compared with DeBERTa + LSTM and DeBERTa + BiLSTM, OnDeBERTa achieves the best performance, with Accuracy, Precision, Recall, and F1-score reaching 94.18%, 92.26%, 91.02%, and 91.64%, respectively. Since all decoder-based models use a one-layer decoder with attention pooling for fair comparison, the improvement is moderate but consistent across all four metrics. These results suggest that the ordered-neuron memory mechanism of ON-LSTM is more effective than standard LSTM and BiLSTM in capturing negation, contrastive structures, and long-distance sentiment cues in e-commerce reviews.

D. Ablation Experiment

To disentangle the effects of pooling and decoder depth, we ablate OnDeBERTa by varying the ON-LSTM decoder depth (1 vs 2 layers) and the pooling strategy (Last, Mean, Attention), while keeping all other settings fixed as in Section 4.2. The results are shown in Table II.

TABLE II. IMPACT OF POOLING AND ON-LSTM DEPTH ON ONDEBERTA

Decoder Depth	Pooling	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
1 layer	Last	92.86	90.22	88.70	89.45
1 layer	Mean	93.42	91.02	89.58	90.29
1 layer	**Attention**	**94.18**	**92.26**	**91.02**	**91.64**
2 layers	Last	93.98	91.48	89.95	90.71
2 layers	Mean	94.55	92.30	90.92	91.60
2 layers	**Attention**	**95.30**	**93.45**	**92.22**	**92.83**

Table II shows that Attention Pooling consistently achieves the best performance under both ON-LSTM depths, indicating its effectiveness in aggregating key sentiment information from review representations. With one ON-LSTM layer, Attention Pooling obtains 94.18% Accuracy and 91.64% F1-score, outperforming Last and Mean Pooling. With two ON-LSTM layers, it further improves to 95.30% Accuracy and 92.83% F1-score.

The results also show that increasing the ON-LSTM depth improves all pooling strategies. The F1-score increases from 89.45% to 90.71% for Last Pooling, from 90.29% to 91.60% for Mean Pooling, and from 91.64% to 92.83% for Attention Pooling. This trend verifies the effectiveness of ON-LSTM hierarchical memory modeling. In this study, the one-layer model is used for the main fair comparison, while the two-layer variant demonstrates the benefit of deeper ON-LSTM decoding.

To further illustrate the effect of the proposed model, we select a negative review example: "The product looked good at first, and the package arrived on time. However, after only two days of use, it stopped working and the customer service was not helpful." As shown in Fig. 4, DeBERTa-base tends to distribute attention over local sentiment words such as "good," "arrived," "working," and "helpful." In contrast, OnDeBERTa places stronger attention on the contrastive cue "However" and the decisive negative fragments "stopped working" and "not helpful." This indicates that OnDeBERTa can better capture contrastive structures and delayed negative evidence in complex e-commerce reviews.

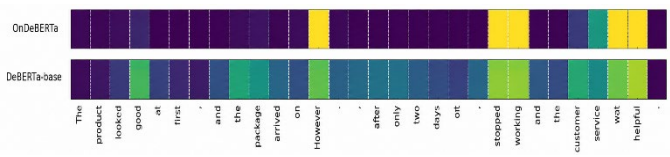


Fig. 4. Token-level attention weight heatmap: DeBERTa-base vs OnDeBERTa

V. CONCLUSION

This paper proposed OnDeBERTa, an enhanced DeBERTa-based model for e-commerce review sentiment classification. By integrating an ON-LSTM decoder and attention pooling after the DeBERTa encoder, the model strengthens hierarchical sequence modeling and improves the capture of negation, contrastive expressions, and long-distance sentiment dependencies. Experiments on the Amazon Review Polarity dataset show that the one-layer OnDeBERTa achieves 94.18% Accuracy, 92.26% Precision, 91.02% Recall, and 91.64% F1-score, outperforming DeBERTa-base, DeBERTa + LSTM, and DeBERTa + BiLSTM under the same decoder depth.

The ablation results further show that Attention Pooling consistently outperforms Last Pooling and Mean Pooling, indicating its effectiveness in aggregating key sentiment information. Increasing the ON-LSTM depth also improves all pooling settings, which verifies the contribution of hierarchical memory modeling. In this study, the one-layer model is used as the main setting for fair comparison, while the two-layer variant demonstrates the potential benefit of deeper ON-LSTM decoding. In addition, the attention visualization case study shows that OnDeBERTa can focus more accurately on contrastive cues and decisive negative fragments than DeBERTa-base. Future work will extend the model to multilingual e-commerce reviews, aspect-based sentiment analysis, and lightweight deployment scenarios [12].

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